

LVT example

How to configure and run LVT

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LVT config file

```
#Overall driver options
LVT running mode:                          "Data intercomparison"
Map projection of the LVT analysis:        latlon
LVT output format:                         netcdf
LVT output methodology:                    "2d gridspace"
LVT output restart files:                  0

Analysis data sources:  "LIS output"  "GLEAM"

Start mode:                                coldstart
LVT restart output interval:               1mo
LVT restart filename:                      none
Starting year:                             2018
Starting month:                               1
Starting day:                                 1
Starting hour:                                0
Starting minute:                              0
Starting second:                              0
Ending year:                               2021
Ending month:                                 1
Ending day:                                   1
Ending hour:                                  0
Ending minute:                                0
Ending second:                                0
LVT clock timestep:                        1da
```

- We are using the LVT mode “data intercomparison”
- We are using two data sources (or datastreams):
  - Our LIS output and GLEAM
- We will do a 3-year comparison (from 1 Jan 2018 to 1 Jan 2021)
- Daily LVT timestep

LVT config file

```
#LaPlata domain
Run domain lower left lat:                -38.98
Run domain lower left lon:                -74.98
Run domain upper right lat:               -13.02
Run domain upper right lon:               -40.02
Run domain resolution (dx):                 0.04
Run domain resolution (dy):                 0.04
Input domain and mask data file:        ./lis_input.laplata.noahmp401.4km.nc

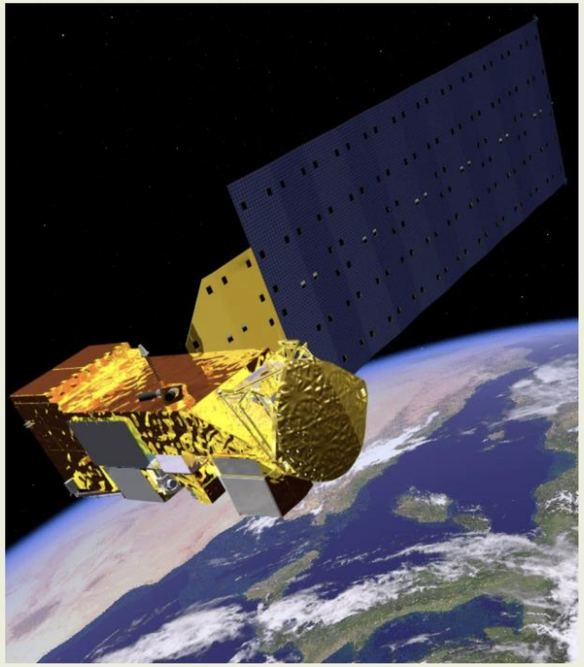
#                    Datastream 1                    |                    Datastream 2
# Short   Selected                  Time   Total |   Short   Selected                  Time   Total
#  Name     Levels  Units    Dir  Avg  Levels |    Name     Levels  Units    Dir  Avg  Levels
LVT datastream attributes table::
Qle             1 1  W/m2     UP    1     1       Qle             1 1  W/m2     UP    1     1
::
```

- We will perform our analysis over the entire domain for our LIS simulation; however, we could specify a sub-set of this domain
- We will compare the latent heat flux (Qle) from Datastream 1 (the LIS output) against Datastream 2 (the GLEAM reference dataset)

The GLEAM dataset

Method

Global Land Evaporation Amsterdam Model



General

The **Global Land Evaporation Amsterdam Model (GLEAM)** is a set of algorithms that separately estimate the different components of land evaporation (often referred to as '*evapotranspiration*'): transpiration, bare-soil evaporation, interception loss, open-water evaporation and sublimation. Additionally, GLEAM provides surface and root-zone soil moisture, potential evaporation and evaporative stress conditions

The rationale of the method is to maximize the recovery of information on evaporation contained in current satellite observations of climatic and environmental variables

- GLEAM (Global Land Evaporation Amsterdam Model) is a reference dataset that provides estimates of total/components of land evaporation, soil moisture, and other variables
- Data is available from: https://www.gleam.eu/

LVT config file

```
#Analysis options
Apply external mask:                    0
Temporal (monthly) mask flags:          1 1 1 1 1 1 1 1 1 1 1 1
External mask directory:                none
Compute ensemble metrics:               0
Compute information theory metrics:     0
Metrics attributes file:                ./laplata/hymap/GLEAM/METRICS_GLEAM.TBL
Observation count threshold:            0
Metrics computation frequency:          1da
Temporal lag in metric computations:    0da
Spatial averaging mode:                 "pixel-by-pixel"
Regional mask file for spatial averaging: none

Metrics output directory:               ./laplata/hymap/GLEAM/OL
Metrics output frequency:               1da
Apply temporal smoothing to obs:        0
Obs temporal smoothing window half length: none
Obs temporal smoothing window interval:    none
Time series location file:              ./laplata/hymap/GLEAM/TS_LOCATIONS_GLEAM.TBL
Variable-based stratification:          0
Stratification variable:                none
Stratification threshold:               0
Confidence interval (%):                95
External data-based stratification:     0
Stratification attributes file:         none
Seasonal cycle interval type:           monthly
Seasonal cycle minimum count threshold: 0
Average diurnal cycle minimum count threshold:    0
Averaging window for computing mean values in anomaly calculations: monthly
```

- Calculate these metrics using LVT
- Calculate the metrics daily
- Do the calculation on each pixel (grid point)
- Output the metrics daily
- If desired, calculate a time-series as specified

LVT config file

```
#Datastream sources
LIS output analysis data class:                 LSM
LIS output number of surface model types:       2
LIS output surface model types:                 "LSM"  "Openwater"
LIS output model name:                          "Noah-MP.4.0.1"
LIS output domain and parameter file:           ./lis_input.laplata.noahmp401.4km.nc
LIS output directory:                           ../lis/laplata/hymap/OL
LIS output naming style:                        "3 level hierarchy"
LIS output methodology:                         "2d gridspace"
LIS output format:                              netcdf
LIS output interval:                            1da
LIS output attributes file:                     ../lis/tables/NOAHMP401_OUTPUT_LIST.TBL.dyn.full

#GLEAM observations
GLEAM data directory:                            ./input/RS_DATA/GLEAM
GLEAM data version:                              "3.3a"
```

- Point to our two datastreams (our LIS “OL” output and the GLEAM data)
- The “LIS output” settings are the same as were specified in the lis.config file that we used to generate the LIS output

The METRICS file

#name	Use	Time	WriteTS	ExtractTS	Threshold(min/max)		ASC	ADC	short_name
Mean:	1	1	0	1	-9999.0	-9999.0	0	0	#MEAN
Anomaly:	0	0	0	0	-9999.0	-9999.0	0	0	#Anomaly
Min:	0	0	0	0	-9999.0	-9999.0	0	0	#MIN
Max:	0	0	0	0	-9999.0	-9999.0	0	0	#MAX
MinTime:	0	0	0	0	-9999.0	-9999.0	0	0	#MINTIME
MaxTime:	0	0	0	0	-9999.0	-9999.0	0	0	#MAXTIME
Sum:	0	0	0	0	-9999.0	-9999.0	0	0	#SUM
Standard deviation:	0	0	0	0	-9999.0	-9999.0	0	0	#STDEV
Variance:	0	0	0	0	-9999.0	-9999.0	0	0	#VARIANCE
Bias:	1	1	0	1	-9999.0	-9999.0	0	0	#BIAS
RMSE:	1	1	0	1	-9999.0	-9999.0	0	0	#RMSE
ubRMSE:	0	0	0	0	-9999.0	-9999.0	0	0	#ubRMSE
Anomaly RMSE:	0	0	0	0	-9999.0	-9999.0	0	0	#ARMSE
Mean absolute error:	0	0	0	0	-9999.0	-9999.0	0	0	#MAE
Anomaly correlation:	0	0	0	0	-9999.0	-9999.0	0	0	#ACORR
Raw correlation:	0	0	0	0	-9999.0	-9999.0	0	0	#RCORR
Rank correlation:	0	0	0	0	-9999.0	-9999.0	0	0	#RNKCORR
Probability of detection (PODy):	0	0	0	0	0.1	0.3	0	0	#PODY
Probability of detection (PODn):	0	0	0	0	0.1	0.3	0	0	#PODN
False alarm ratio (FAR):	0	0	0	0	0.1	0.3	0	0	#FAR

- We will calculate the Mean, Bias, and RMSE metrics at each 1day interval
- We will NOT write netCDF files each day (“WriteTS”)
- Write to ASCII/text files at the time series locations each day (“ExtractTS”)

The TS_LOCATIONS file

```
#Number of stations
1
#Location style (1-lat/lon, 2-col/row, 3-tile, 4-polygon, 5-map)
1
#Name (then, next line), min frac
SAMER
-38.93  -74.98  -13.02  -40.02  0.01
```

- We will calculate a time series (area-averaged) over this lat-lon box
- In this case, I have set this box as the entire domain
- A user could set any number of time-series; here, we are looking at just “1”
- A user could set these as individual grid boxes, by making the north and south latitude the same value, and the west and east longitude the same.

Running LVT

- To run LVT, on the command line, a user would type this command:

```
./LVT lvt.config.laplata.noahmp401.hymap.GLEAM.OL
```

- We can run LVT again, this time using an lvt.config file that points to the LIS output from our “DA” experiment instead:

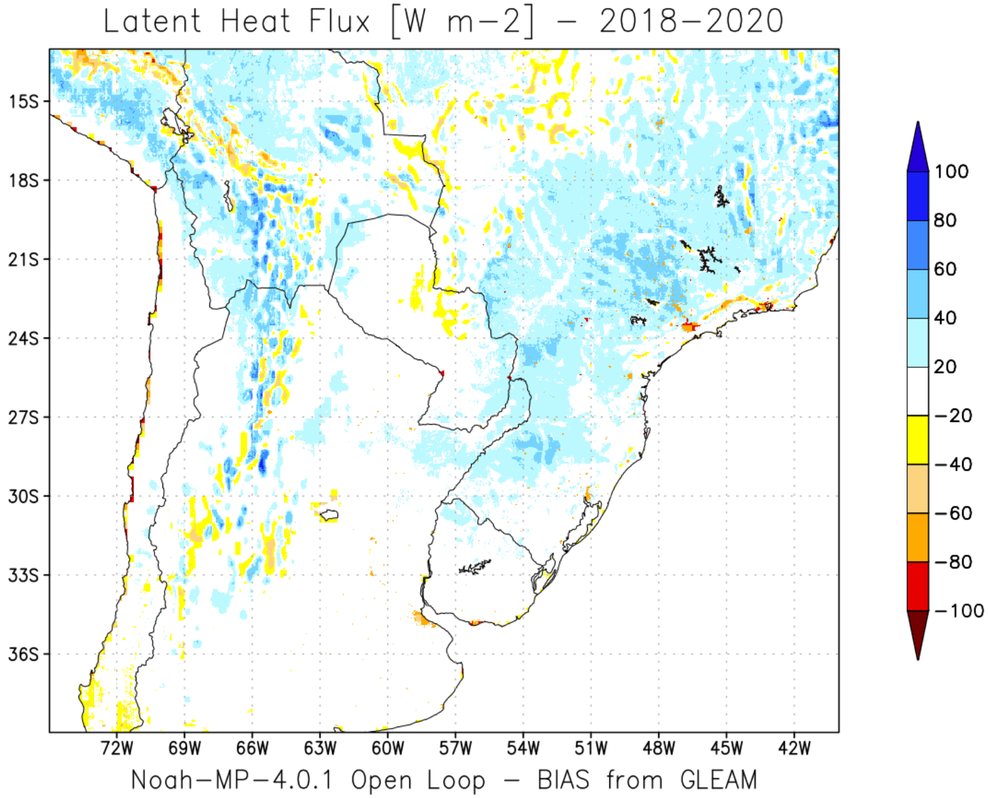
```
./LVT lvt.config.laplata.noahmp401.hymap.GLEAM.DA
```

- These two lvt.config files are essentially identical, except for:

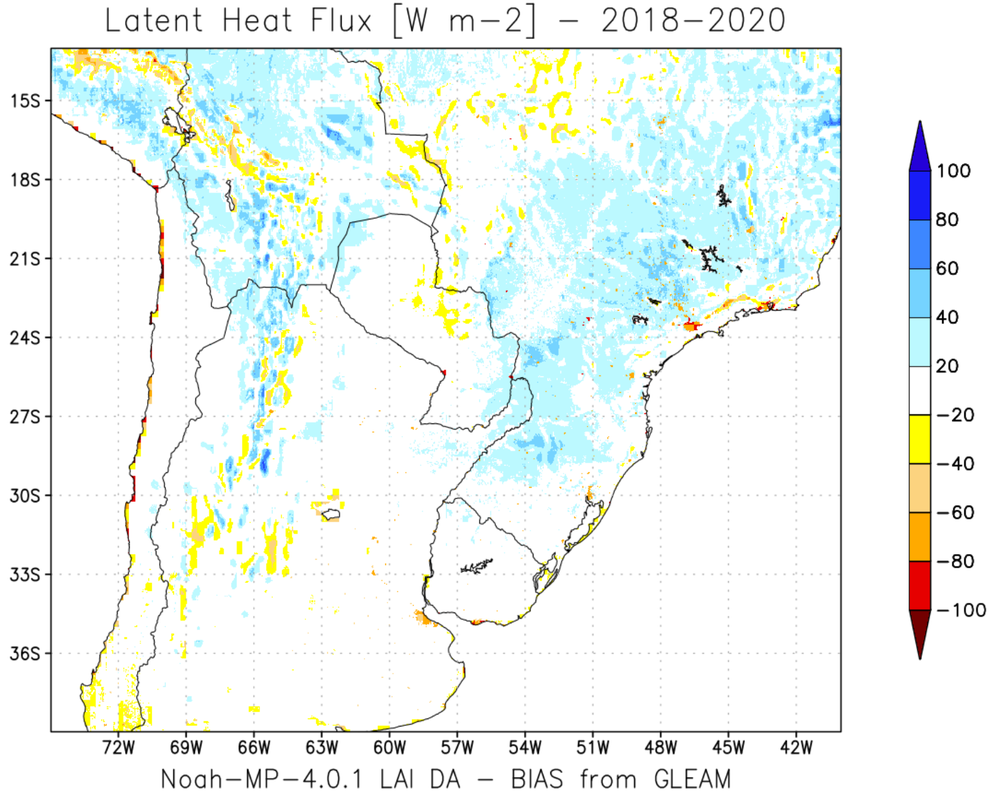
1. We point to different LIS output directories (OL or DA)
2. We change the LVT output directory and diagnostic file output to different locations, so we have the output separated and don’t overwrite our previous LVT output

Example: LVT output of BIAS from GLEAM

Open Loop:



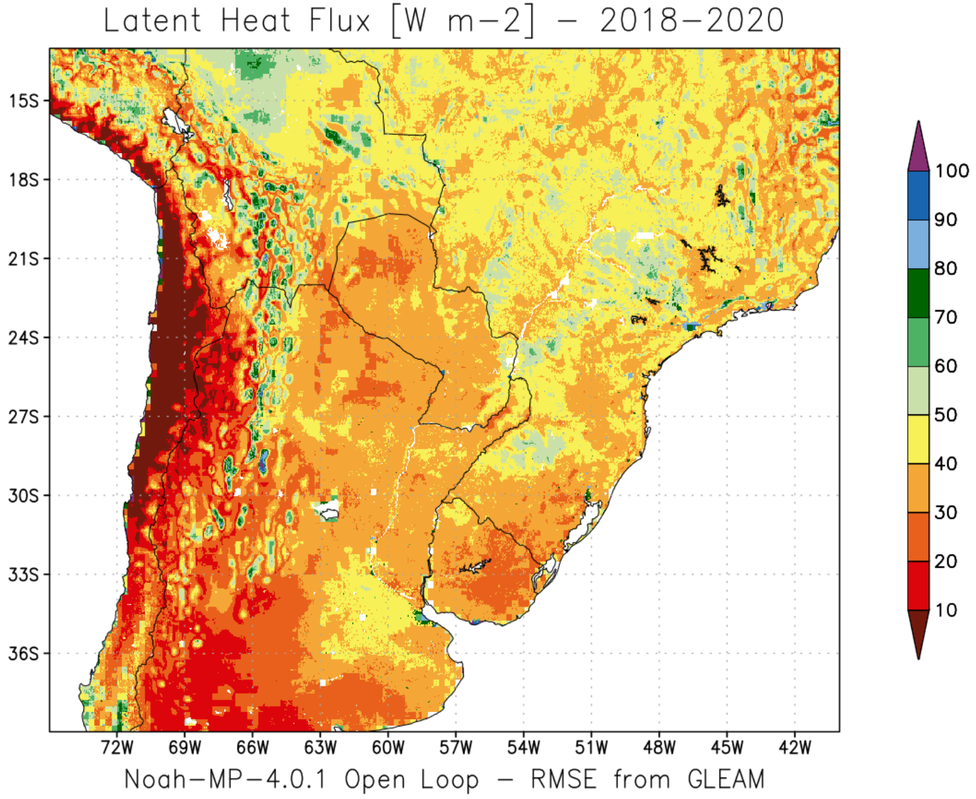
LAI DA:



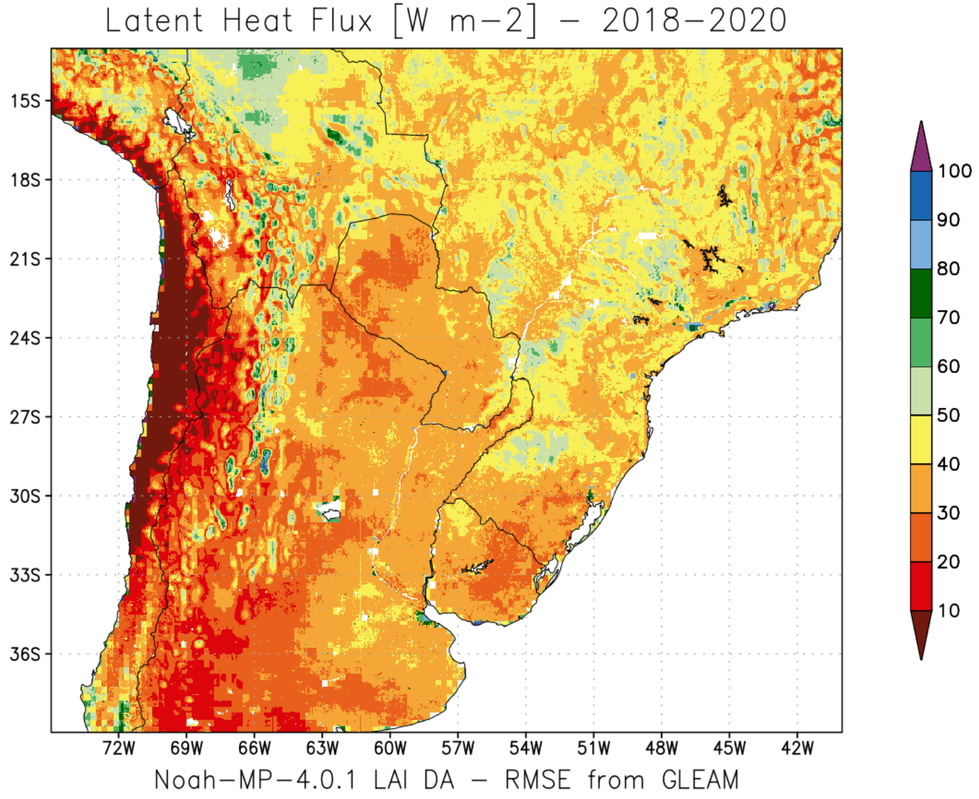
- Data assimilation of LAI (right figure) reduced the BIAS against the GLEAM reference Latent Heat Flux (Qle)

Example: LVT output of RMSE from GLEAM

Open Loop:

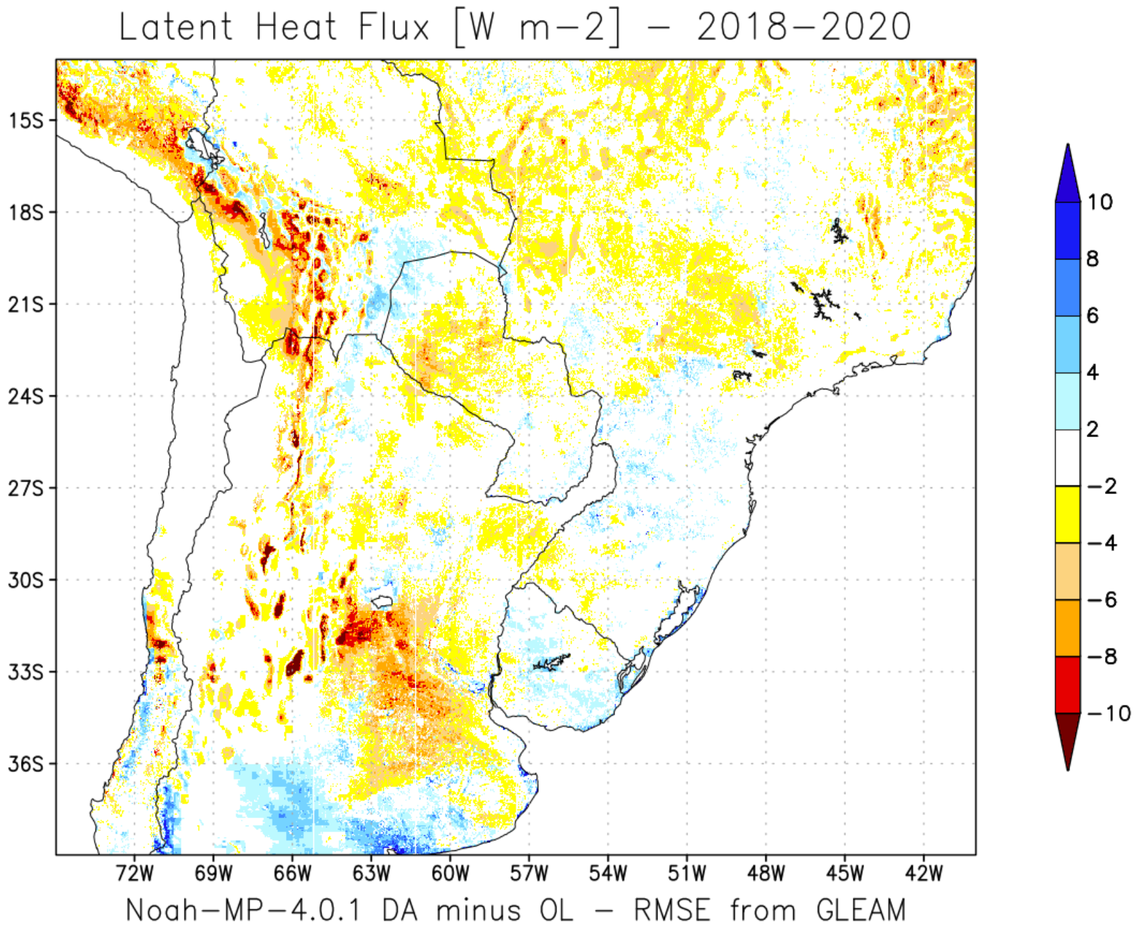


LAI DA:



- Data assimilation of LAI (right figure) reduced the RMSE against the GLEAM reference Latent Heat Flux (Qle)
- RMSE metric = Root-Mean-Square Error

Example: LVT output of RMSE from GLEAM



- We can also plot the DIFFERENCE between the two previous figures
- Shown here is the [RMSE of the DA simulation minus the RMSE of the OL simulation], or:

  [RMSE DA] – [RMSE OL]

- The negative values (warm/red colors) show areas where LAI DA *reduced the RMSE* as compared to GLEAM reference latent heat flux